Muhammad Muizz Zafar

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Software Engineer | Data Platforms | Cloud Infrastructure

Skills

Programming Languages:	Java, Python, GoLang, JavaScript, Kotlin, C#, SQL, HTML/CSS, PHP, Flutter, C++
Tools/Frameworks:	AWS, Splunk, Flask, React, Vue, Angular, GraphQl, Docker, Git, Firebase, Linux

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Software Developer — Jan 2025 - Present
Elections Canada — Ottawa, ON

- Designed and implemented a scalable **ETL validation framework** using **Python** and **Azure DevOps CI/CD pipelines** to support automated validation of large-scale data pipelines across multiple environments.
- Developed automated **data validation and scoring pipelines** to monitor ETL reliability and ensure data consistency across **Oracle and SQL Server systems processing high-volume datasets**.
- Built internal **data analytics dashboards** using **Python, Streamlit, and Plotly Dash** to visualize operational metrics and support data quality monitoring for production systems.
- Coordinated version-controlled database deployments and release processes, improving traceability and reliability of **production data infrastructure**.
- Collaborated in an **Agile cross-functional team** with cloud engineers and DevOps teams to improve **distributed data workflows** and system reliability.

Full Stack Developer Intern — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Redesigned the University of Ottawa's emergency alert website with **C#** and **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily. **website: ue.uottawa.ca/**

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring systems using **AWS Lambda** to analyze uploaded client files and images, supporting automated vulnerability scanning workflows in a cloud-native environment.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve error handling, reliability, and execution efficiency of distributed cloud workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks across large volumes of cloud artifacts.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to trigger metadata updates for incoming cloud data events, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring to track system logs and failures, improving observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to validate cloud services and ensure reliability of distributed serverless systems.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Utilized containerization concepts to create **Docker** containers for the full- stack application for PROD deployment
- Automated the event detection process by adding events to **PostgreSQL** DB using **Python** and **GraphQl** which helped reduce testing time by **10 seconds**

Full Stack Developer Intern — May 2021 - August 2021
University of Ottawa — Ottawa, ON

- Optimized existing **C#** REST APIs by reducing **SQL** query execution time for event data lookup by **30 %** and lines of code by **40 %**
- Revamped the event registration and admin webpages by using **Angular** material UI components to improve UI layout and reducing redundant lines of code by almost **50 %**